

Elouan Salaun

Data Scientist | Machine Learning Engineer

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PROFILE

Data Scientist / Machine Learning Engineer with experience building end-to-end ML systems from data extraction and NLP to predictive modeling, cloud deployment, and business adoption. Built AI decision-support tools using Python, SQL, XGBoost, NLP/RAG, and Azure, including a pricing model that increased simulated bid win rate from 15% to 40%.

EXPERIENCE

Data Scientist - Research Internship/MSc.A Partnership | Commissionnaires du Québec

Montréal, 09/2024 – 09/2026

Security-services organization bidding on public tenders across Québec. Sole owner of the data & AI workstream, from scoping to deployment.

- Designed and shipped a complete AI-driven tool covering the full RFP response process; presented multiple times to the executive committee and to all staff.
- Built a pricing model (XGBoost / Learning-to-Rank) combining market dynamics with contract-specific technical features: simulated win rate of 40% vs. 15% for the pricing rule in use, +10% profits against historical results on retrospective backtests.
- Consolidated the data foundation from scratch (internal spreadsheets and PDFs plus external APIs and scraping) into a single contract database; extraction from contractual documents via NLP and hierarchical RAG.
- Developed a similar-contract retrieval model (neural network with human feedback) to surface comparable past bids for any new tender.
- Delivered the decision-support application (Streamlit) and deployed the stack on Azure: ADLS Gen2 data lake, Functions, AppService, Containers.
- Built automated Excel spreadsheets with macros (VBA) for reporting outside the tool.

Vice-President of the School's Informatics & History Associations | ENSAM

France, 2023-2024

- Designed and programmed an ESP32 real-time sales-recording device for a 3,000-person event: MQTT transmission, centralized inventory and demand forecasting with restocking anticipation across points of sale.
- Created an annotated virtual-tour website of the school; awarded a €1,000 FSDIE grant.

Summer Internship – Industrial Engineer | Blanchisserie d’Armor

France, Summer 2023

- Regional industrial laundry with 230 individual clients and 180T/week.
- Drafted a report on process optimization; applying the Traveling Salesman Problem to a delivery truck fleet.

PUBLICATION

Leveraging artificial intelligence and strategic data for tender response processes

CENTERIS 2026

E. Salaun, M. Belmouadden, R. Pellerin, A. Neumann – conference paper, Procedia Computer Science.

PROJECTS

SEAO Public Procurement MCP Connector for Claude/GPT

2026

- Local MCP server/tool that lets an LLM interrogate Québec’s SEAO open procurement data in plain language: ingestion pipeline from raw publications, cleaning and normalization into SQLite, hybrid query core, concurrent analysis inside the model’s reasoning loop. *(Python, SQL, MCP, LLM)*

Optimal surveillance of a site from airborne LiDAR

2026

- End-to-end system using 3D LiDAR data to optimize sensors placement for site surveillance, combining exact visibility modeling, mathematical optimization, adversarial intruder simulation, and deep-learning surrogate models. *(Python, MILP, NSGA-II, U-net)*

LiDAR AI Agent

2026

- An agentic AI system that converts natural-language questions into deterministic LiDAR→ML→GIS workflows using LangGraph and Claude. Developed a Random Forest point-cloud classifier and automated building extraction/analysis pipeline, with validation, caching, and interactive geospatial visualization. *(Python, LangGraph, Claude, RandomForest, GeoPandas)*

Jet fighter flight simulation and Reinforcement Learning

2025

- Developed a physics-based 3/6-DOF F-16 simulator from NASA aerodynamic data and trained PPO reinforcement-learning agents in a Gymnasium environment to perform flight recovery, navigation, sustained turns, and aerobatic maneuvers. *(Python, simulation, PPO RL)*

Investment Portfolio Optimization

2025

- Liability-driven portfolio construction: Markowitz frontier and GMV under estimation-error sensitivity, Gaussian and Cornish-Fisher VaR/CVaR, CPPI with explicit drawdown constraint, geometric Brownian motion Monte Carlo, CIR rates and bond pricing. *(Python, statistics)*

Behavioral Analysis of a Superalloy Safran Aircraft Engines – semester project	2024
■ Instrumented a forging process, then identified the constitutive law from the collected data using machine learning; validated against numerical simulations. <i>(Python, ML, simulations)</i>	
Rocket Nose Cone Manufacturing by Filament Winding CNES (Perseus) – semester project	2024
■ Determined filament lay-down trajectories as geodesics, programmed the 4-axis winding machine (G-code), and built an interface to compile, verify and visualize the trajectories. <i>(Python, mathematics)</i>	

EDUCATION

Research Master's (MSc.A), Industrial Engineering Polytechnique Montréal	Montréal, 2024-2026
Thesis: <i>Tender response process: integrating artificial intelligence methods that leverage contractual and competitive data.</i>	
Supervised by Robert Pellerin and Anas Neumann	
Member of: Jarislowsky/AtkinsRéalis research chair & CIRRELT (Interuniversity Research Centre on Enterprise Networks, Logistics and Transportation)	
Standing: GPA 4.0/4.0 – Distinction (Honors)	
Engineering degree (Master’s level), Grande École Program Arts et Métiers ParisTech (ENSAM)	France, 2022-2026
A leading generalist engineering school in France.	
Standing: Top 10% – Recipient of an international study scholarship from the alumni society	
Quantitative core: applied mathematics (Fourier analysis, symbolic and numerical computation, optimization), programming and databases, automation and control, embedded systems.	
Engineering core: mechanical and process engineering, materials, energy, industrial engineering (production management, Lean, Six Sigma), economics and accounting.	
Preparatory Classes for the Grandes Écoles (CPGE), PTSI/PT* Lycée Livet	France, 2020-2022
Preparation for the entrance exams for top French engineering schools.	
Weekly coursework: 9h mathematics, 8.5h engineering science, 8h physics, 2h computer science, 4h languages, 4h written exams, 2h oral exams	
SKILLS	
General: Problem solving, teamwork, autonomy, communication, rigor	
Programming: Python (pandas, NumPy, scikit-learn, XGBoost, PyTorch), SQL, VBA, MATLAB, G-code, C/C++ (embedded)	
Machine Learning: Supervised learning and gradient boosting, Learning-to-Rank, neural networks with human feedback, NLP and information extraction, time series, unsupervised learning, testing	
LLM systems: Retrieval-augmented generation (hierarchical RAG), Model Context Protocol (MCP) servers, local LLM deployment, prompt and context engineering	
Data & cloud: Azure (ADLS Gen2, Functions, App Service, Containers), SQLite, Docker, Git, Streamlit, API integration and web scraping, Excel/VBA automation	
Operations research: Mathematical optimization, convex and stochastic optimization, Monte Carlo simulation, numerical simulation	
Languages: French (native), English (fluent, C1 - TOEIC 950)	
Certifications: IBM Data Science Specialization (160h, Coursera), Stanford Machine Learning Specialization (80h, Coursera)	